Code

```python
import pygame
from pygame import mixer
pygame.init()

black = (0, 0, 0)
white = (255, 255, 255)
gray = (128, 128, 128)
dark_gray = (50, 50, 50)
light_gray = (170, 170, 170)
blue = (0, 255, 255)
red = (255, 0, 0)
green = (0, 255, 0)
gold = (212, 175, 55)
WIDTH = 1400
HEIGHT = 800
active_length = 0
active_beat = 0

# sounds
'''
hi_hat = mixer.Sound('sounds\kit2\hi hat.wav')
snare = mixer.Sound('sounds\kit2\snare.wav')
kick = mixer.Sound('sounds\kit2\kick.wav')
crash = mixer.Sound('sounds\kit2\crash.wav')
clap = mixer.Sound('sounds\kit2\clap.wav')
tom = mixer.Sound("sounds\kit2\\tom.wav")
'''
hi_hat = mixer.Sound('sounds\hi hat.wav')
snare = mixer.Sound('sounds\snare.wav')
kick = mixer.Sound('sounds\kick.wav')
crash = mixer.Sound('sounds\crash.wav')
clap = mixer.Sound('sounds\clap.wav')
tom = mixer.Sound("sounds\\tom.wav")

screen = pygame.display.set_mode([WIDTH, HEIGHT])
pygame.display.set_caption('The Beat Maker')
label_font = pygame.font.Font('Roboto-Bold.ttf', 32)
```

```
38  medium_font = pygame.font.Font('Roboto-Bold.ttf', 24)
39  beat_changed = True
40  timer = pygame.time.Clock()
41  fps = 60
42  beats = 8
43  bpm = 240
44  instruments = 6
45  playing = True
46  clicked = [[-1 for _ in range(beats)] for _ in range(instruments)]
47  active_list = [1 for _ in range(instruments)]
48  pygame.mixer.set_num_channels(instruments * 3)
49  save_menu = False
50  load_menu = False
51  saved_beats = []
52  file = open('saved_beats.txt', 'r')
53  for line in file:
54      saved_beats.append(line)
55  beat_name = ''
56  typing = False
57  index = 100
58
59
60  def draw_grid(clicks, beat, actives):
61      boxes = []
62      left_box = pygame.draw.rect(screen, gray, [0, 0, 200, HEIGHT - 200], 5)
63      bottom_box = pygame.draw.rect(screen, gray, [0, HEIGHT - 200, WIDTH, 200], 5)
64      for i in range(instruments + 1):
65          pygame.draw.line(screen, gray, (0, i * 100), (200, i * 100), 3)
66      colors = [gray, white, gray]
67      hi_hat_text = label_font.render('Hi Hat', True, colors[actives[0]])
68      screen.blit(hi_hat_text, (30, 30))
69      snare_text = label_font.render('Snare', True, colors[actives[1]])
70      screen.blit(snare_text, (30, 130))
71      kick_text = label_font.render('Bass Drum', True, colors[actives[2]])
72      screen.blit(kick_text, (30, 230))
73      crash_text = label_font.render('Crash', True, colors[actives[3]])
74      screen.blit(crash_text, (30, 330))
```

```
    clap_text = label_font.render('Clap', True, colors[actives[4]])
    screen.blit(clap_text, (30, 430))
    tom_text = label_font.render('Floor Tom', True, colors[actives[5]])
    screen.blit(tom_text, (30, 530))
    for i in range(beats):
        for j in range(instruments):
            if clicks[j][i] == -1:
                color = gray
            else:
                if actives[j] == 1:
                    color = green
                else:
                    color = dark_gray
            rect = pygame.draw.rect(screen, color,
                                    [i * ((WIDTH - 200) // beats) + 205, (j * 100) + 5, ((WIDTH - 200) // beats) - 10,
                                     90], 0, 3)
            pygame.draw.rect(screen, gold, [i * ((WIDTH - 200) // beats) + 200, j * 100, ((WIDTH - 200) // beats), 100],
                             5, 5)
            pygame.draw.rect(screen, black,
                             [i * ((WIDTH - 200) // beats) + 200, j * 100, ((WIDTH - 200) // beats), 100],
                             2, 5)
            boxes.append((rect, (i, j)))
    active = pygame.draw.rect(screen, blue,
                              [beat * ((WIDTH - 200) // beats) + 200, 0, ((WIDTH - 200) // beats), instruments * 100],
                              5, 3)
    return boxes


def play_notes():
    for i in range(len(clicked)):
        if clicked[i][active_beat] == 1 and active_list[i] == 1:
            if i == 0:
                hi_hat.play()
            if i == 1:
                snare.play()
            if i == 2:
```

```
                kick.play()
            if i == 3:
                crash.play()
            if i == 4:
                clap.play()
            if i == 5:
                tom.play()


def draw_save_menu(beat_name, typing):
    pygame.draw.rect(screen, black, [0, 0, WIDTH, HEIGHT])
    menu_text = label_font.render('SAVE MENU: Enter a Name for this beat', True, white)
    screen.blit(menu_text, (400, 40))
    exit_btn = pygame.draw.rect(screen, gray, [WIDTH - 200, HEIGHT - 100, 180, 90], 0, 5)
    exit_text = label_font.render('Close', True, white)
    screen.blit(exit_text, (WIDTH - 160, HEIGHT - 70))
    saving_btn = pygame.draw.rect(screen, gray, [WIDTH // 2 - 100, HEIGHT * 0.75, 200, 100], 0, 5)
    saving_text = label_font.render('Save Beat', True, white)
    screen.blit(saving_text, (WIDTH // 2 - 70, HEIGHT * 0.75 + 30))
    if typing:
        pygame.draw.rect(screen, dark_gray, [400, 200, 600, 200], 0, 5)
    entry_rect = pygame.draw.rect(screen, gray, [400, 200, 600, 200], 5, 5)
    entry_text = label_font.render(f'{beat_name}', True, white)
    screen.blit(entry_text, (430, 250))
    return exit_btn, saving_btn, beat_name, entry_rect


def draw_load_menu(index):
    loaded_clicked = []
    loaded_beats = 0
    loaded_bpm = 0
    pygame.draw.rect(screen, black, [0, 0, WIDTH, HEIGHT])
    menu_text = label_font.render('LOAD MENU: Select a beat to load in', True, white)
    screen.blit(menu_text, (400, 40))
    exit_btn = pygame.draw.rect(screen, gray, [WIDTH - 200, HEIGHT - 100, 180, 90], 0, 5)
    exit_text = label_font.render('Close', True, white)
```

```
    screen.blit(exit_text, (WIDTH - 160, HEIGHT - 70))
    loading_btn = pygame.draw.rect(screen, gray, [WIDTH // 2 - 100, HEIGHT * 0.87, 200, 100], 0, 5)
    loading_text = label_font.render('Load Beat', True, white)
    screen.blit(loading_text, (WIDTH // 2 - 70, HEIGHT * 0.87 + 30))
    delete_btn = pygame.draw.rect(screen, gray, [WIDTH // 2 - 400, HEIGHT * 0.87, 200, 100], 0, 5)
    delete_text = label_font.render('Delete Beat', True, white)
    screen.blit(delete_text, (WIDTH // 2 - 385, HEIGHT * 0.87 + 30))
    if 0 <= index < len(saved_beats):
        pygame.draw.rect(screen, light_gray, [190, 100 + index*50, 1000, 50])
    for beat in range(len(saved_beats)):
        if beat < 10:
            beat_clicked = []
            row_text = medium_font.render(f'{beat + 1}', True, white)
            screen.blit(row_text, (200, 100 + beat * 50))
            name_index_start = saved_beats[beat].index('name: ') + 6
            name_index_end = saved_beats[beat].index(', beats:')
            name_text = medium_font.render(saved_beats[beat][name_index_start:name_index_end], True, white)
            screen.blit(name_text, (240, 100 + beat * 50))
        if 0 <= index < len(saved_beats) and beat == index:
            beats_index_end = saved_beats[beat].index(', bpm:')
            loaded_beats = int(saved_beats[beat][name_index_end + 8:beats_index_end])
            bpm_index_end = saved_beats[beat].index(', selected:')
            loaded_bpm = int(saved_beats[beat][beats_index_end + 6:bpm_index_end])
            loaded_clicks_string = saved_beats[beat][bpm_index_end + 14: -3]
            loaded_clicks_rows = list(loaded_clicks_string.split("], ["))
            for row in range(len(loaded_clicks_rows)):
                loaded_clicks_row = (loaded_clicks_rows[row].split(', '))
                for item in range(len(loaded_clicks_row)):
                    if loaded_clicks_row[item] == '1' or loaded_clicks_row[item] == '-1':
                        loaded_clicks_row[item] = int(loaded_clicks_row[item])
                beat_clicked.append(loaded_clicks_row)
                loaded_clicked = beat_clicked
    loaded_info = [loaded_beats, loaded_bpm, loaded_clicked]
    entry_rect = pygame.draw.rect(screen, gray, [190, 90, 1000, 600], 5, 5)
    return exit_btn, loading_btn, entry_rect, delete_btn, loaded_info

```

```python

run = True
while run:
    timer.tick(fps)
    screen.fill(black)
    boxes = draw_grid(clicked, active_beat, active_list)
    # drawing lower menu
    play_pause = pygame.draw.rect(screen, gray, [50, HEIGHT - 150, 200, 100], 0, 5)
    play_text = label_font.render('Play/Pause', True, white)
    screen.blit(play_text, (70, HEIGHT - 130))
    if playing:
        play_text2 = medium_font.render('Playing', True, dark_gray)
    else:
        play_text2 = medium_font.render('Paused', True, dark_gray)
    screen.blit(play_text2, (70, HEIGHT - 100))
    # beats per minute buttons
    bpm_rect = pygame.draw.rect(screen, gray, [300, HEIGHT - 150, 200, 100], 5, 5)
    bpm_text = medium_font.render('Beats Per Minute', True, white)
    screen.blit(bpm_text, (308, HEIGHT - 130))
    bpm_text2 = label_font.render(f'{bpm}', True, white)
    screen.blit(bpm_text2, (370, HEIGHT - 100))
    bpm_add_rect = pygame.draw.rect(screen, gray, [510, HEIGHT - 150, 48, 48], 0, 5)
    bpm_sub_rect = pygame.draw.rect(screen, gray, [510, HEIGHT - 100, 48, 48], 0, 5)
    add_text = medium_font.render('+5', True, white)
    screen.blit(add_text, (520, HEIGHT - 140))
    sub_text = medium_font.render('-5', True, white)
    screen.blit(sub_text, (520, HEIGHT - 90))
    # beats per loop buttons
    beats_rect = pygame.draw.rect(screen, gray, [600, HEIGHT - 150, 200, 100], 5, 5)
    beats_text = medium_font.render('Beats In Loop', True, white)
    screen.blit(beats_text, (612, HEIGHT - 130))
    beats_text2 = label_font.render(f'{beats}', True, white)
    screen.blit(beats_text2, (670, HEIGHT - 100))
    beats_add_rect = pygame.draw.rect(screen, gray, [810, HEIGHT - 150, 48, 48], 0, 5)
    beats_sub_rect = pygame.draw.rect(screen, gray, [810, HEIGHT - 100, 48, 48], 0, 5)
    add_text2 = medium_font.render('+1', True, white)
    screen.blit(add_text2, (820, HEIGHT - 140))
```

(variable) medium_font: Font

```
220         sub_text2 = medium_font.render('-1', True, white)
221         screen.blit(sub_text2, (820, HEIGHT - 90))
222         # clear board button
223         clear = pygame.draw.rect(screen, gray, [1150, HEIGHT - 150, 200, 100], 0, 5)
224         play_text = label_font.render('Clear Board', True, white)
225         screen.blit(play_text, (1160, HEIGHT - 130))
226         # save and load buttons
227         save_button = pygame.draw.rect(screen, gray, [900, HEIGHT - 150, 200, 48], 0, 5)
228         save_text = label_font.render('Save Beat', True, white)
229         screen.blit(save_text, (920, HEIGHT - 140))
230         load_button = pygame.draw.rect(screen, gray, [900, HEIGHT - 98, 200, 48], 0, 5)
231         load_text = label_font.render('Load Beat', True, white)
232         screen.blit(load_text, (920, HEIGHT - 90))
233         # instrument rectangles
234         instrument_rects = []
235         for i in range(instruments):
236             rect = pygame.rect.Rect((0, i * 100), (200, 100))
237             instrument_rects.append(rect)
238         if beat_changed:
239             play_notes()
240             beat_changed = False
241         if save_menu:
242             exit_button, saving_button, beat_name, entry_rect = draw_save_menu(beat_name, typing)
243         elif load_menu:
244             exit_button, loading_button, entry_rect, delete_button, loaded_information = draw_load_menu(index)
245         for event in pygame.event.get():
246             if event.type == pygame.QUIT:
247                 run = False
248             if event.type == pygame.MOUSEBUTTONDOWN and not save_menu and not load_menu:
249                 for i in range(len(boxes)):
250                     if boxes[i][0].collidepoint(event.pos):
251                         coords = boxes[i][1]
252                         clicked[coords[1]][coords[0]] *= -1
253             if event.type == pygame.MOUSEBUTTONUP and not save_menu and not load_menu:
254                 if play_pause.collidepoint(event.pos) and playing:
255                     playing = False
256                 elif play_pause.collidepoint(event.pos) and not playing:
```

```
257                     playing = True
258                     active_beat = 0
259                     active_length = 0
260                 if beats_add_rect.collidepoint(event.pos):
261                     beats += 1
262                     for i in range(len(clicked)):
263                         clicked[i].append(-1)
264                 elif beats_sub_rect.collidepoint(event.pos):
265                     beats -= 1
266                     for i in range(len(clicked)):
267                         clicked[i].pop(-1)
268                 if bpm_add_rect.collidepoint(event.pos):
269                     bpm += 5
270                 elif bpm_sub_rect.collidepoint(event.pos):
271                     bpm -= 5
272                 if clear.collidepoint(event.pos):
273                     clicked = [[-1 for _ in range(beats)] for _ in range(instruments)]
274                 for i in range(len(instrument_rects)):
275                     if instrument_rects[i].collidepoint(event.pos):
276                         active_list[i] *= -1
277                 if save_button.collidepoint(event.pos):
278                     save_menu = True
279                 if load_button.collidepoint(event.pos):
280                     load_menu = True
281                     playing = False
282             elif event.type == pygame.MOUSEBUTTONUP:
283                 if exit_button.collidepoint(event.pos):
284                     save_menu = False
285                     load_menu = False
286                     playing = True
287                     typing = False
288                     beat_name = ''
289                 if entry_rect.collidepoint(event.pos):
290                     if save_menu:
291                         if typing:
292                             typing = False
293                         else:
```

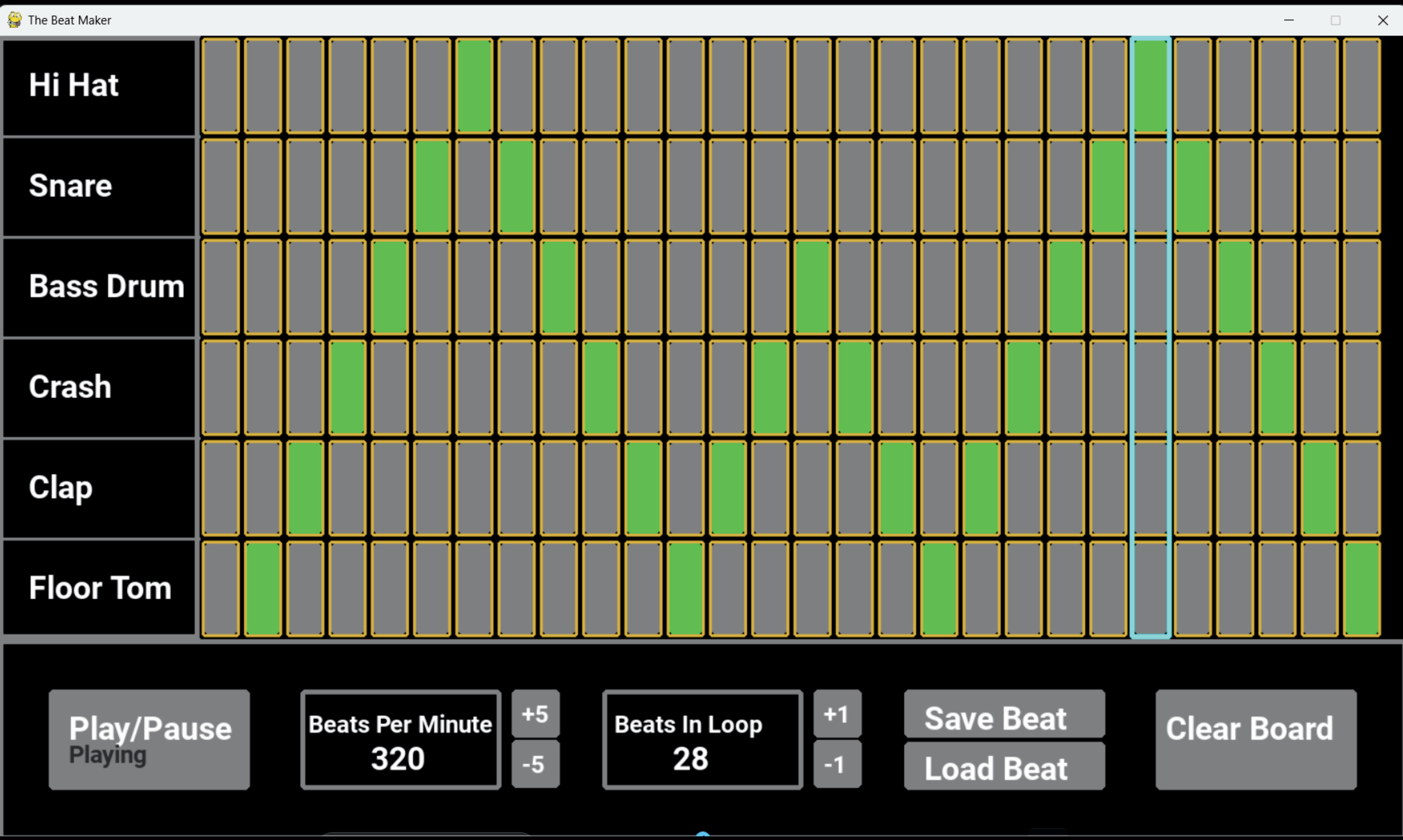
```
                        typing = True
                if load_menu:
                    index = (event.pos[1] - 100) // 50
            if save_menu:
                if saving_button.collidepoint(event.pos):
                    file = open('saved_beats.txt', 'w')
                    saved_beats.append(f'\nname: {beat_name}, beats: {beats}, bpm: {bpm}, selected: {clicked}')
                    for i in range(len(saved_beats)):
                        file.write(str(saved_beats[i]))
                    file.close()
                    save_menu = False
                    load_menu = False
                    playing = True
                    typing = False
                    beat_name = ''
            if load_menu:
                if delete_button.collidepoint(event.pos):
                    if 0 <= index < len(saved_beats):
                        saved_beats.pop(index)
                if loading_button.collidepoint(event.pos):
                    if 0 <= index < len(saved_beats):
                        beats = loaded_information[0]
                        bpm = loaded_information[1]
                        clicked = loaded_information[2]
                        index = 100
                        save_menu = False
                        load_menu = False
                        playing = True
                        typing = False
        if event.type == pygame.TEXTINPUT and typing:
            beat_name += event.text
        if event.type == pygame.KEYDOWN:
            if event.key == pygame.K_BACKSPACE and len(beat_name) > 0:
                beat_name = beat_name[:-1]

    beat_length = 3600 // bpm

```

```
331         if playing:
332             if active_length < beat_length:
333                 active_length += 1
334             else:
335                 active_length = 0
336                 if active_beat < beats - 1:
337                     active_beat += 1
338                     beat_changed = True
339                 else:
340                     active_beat = 0
341                     beat_changed = True
342 
343             pygame.display.flip()
344 
345     file = open('saved_beats.txt', 'w')
346     for i in range(len(saved_beats)):
347         file.write(str(saved_beats[i]))
348     file.close()
349     pygame.quit()
350 
```

Output


The Beat Maker
Hi Hat
Snare
Bass Drum
Crash
Clap
Floor Tom
Play/Pause
Playing
Beats Per Minute
320
+5
-5
Beats In Loop
28
+1
-1
Save Beat
Load Beat
Clear Board

The Beat Maker

SAVE MENU: Enter a Name for this beat

Save Beat

Close

The Beat Maker

LOAD MENU: Select a beat to load in

1	generic rock beat
2	pyramids
3	this is sweet
4	build up
5	rock2
6	this is sweat
7	build up2
8	new song

Delete Beat

Load Beat

Close